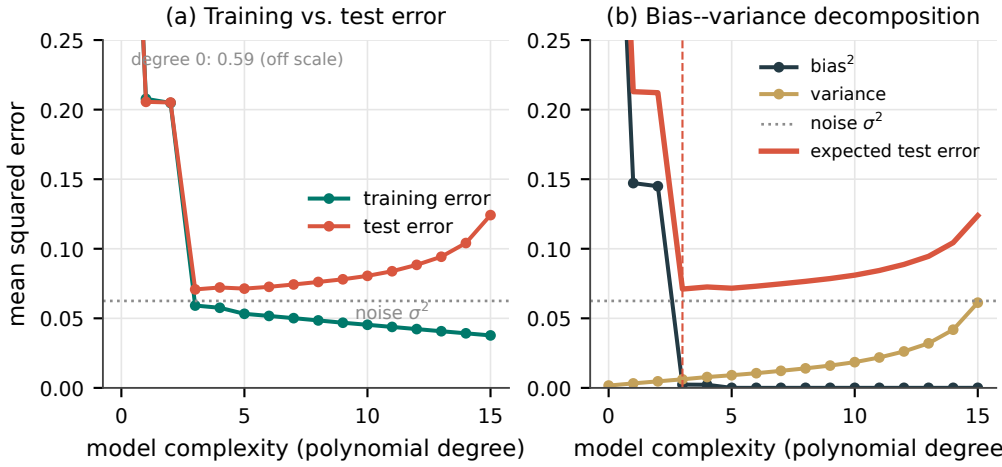
(a) Training vs. test error
(b) Bias--variance decomposition
mean squared error
0.25
0.20
0.15
0.10
0.05
0.00
degree 0: 0.59 (off scale)
training error
test error
noise σ^2
bias^2
variance
noise σ^2
expected test error
0
5
10
15
model complexity (polynomial degree)
model complexity (polynomial degree)